BUSINESS INFORMATION SYSTEMS SECURITY (BISS)

Unit booklet

2023 - 2024 Session

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Unit Aim

To develop students' understanding of the varied security threats and countermeasures in relation to the business environment and the impact on business of security breaches

Unit Learning Outcomes

1. To articulate the security threats in relation to business information systems
2. To analyse the impact on businesses of security breaches
3. Examine the legal, ethical and professional responsibilities in relation to developing secure systems

Unit Overview

This unit concentrates on developing the students' understanding of the value of corporate resources (such as data) and why it is at risk. It looks at the varied security threats to business and the impact this can have on a business or organisation and looks at solutions to these threats. The unit explores the changes in the variety and nature of threats to security due to new technologies being introduces such as the use of mobile devices and cloud computing. Students will be introduced to the importance that technical components such as operating systems, networks and databases have on design of secure solutions as well as application-level security. The contribution of human factors to the security of an organisation's information is covered.

Unit delivery will consist of a lecture and a tutorial each week. The unit will also draw upon case study material, directed learn tasks and where possible guest speakers.

General topic areas covered include: -

1. Corporate information systems, Information systems security, Organisational security, Internal and External threats.
2. Risk assessment processes and methods
3. Network and distributed systems security, intrusion detection systems, monitoring and support systems - risks from the cloud, outsourcing, offshoring and new technologies
4. Applications and Database security
5. Human Factors in security - including mobile devices and the mobile workforce, Social Networking risks (profiling)
6. Cybercrime - identity theft, viruses, Trojans, worms, malware etc
7. Designing secure systems, working processes and practices, and secure interfaces Legal Issues - The Data Protection Act, Computer Misuse Act etc
8. Ethical and Professional Standards - BCS guidelines, ISO standards, Security Policies

Recommended Reading

Hard copy in UoP library

Gibson, D., & Igonor, A. (2015). *Managing risk in information systems*. Jones & Bartlett Learning.

Gobeo, A., Fowler, C., & Buchanan, W. J. (2018). *GDPR and Cyber Security for Business Information Systems*. River Publishers.

Malcolm Harkins (2016) *Managing Risk and Information Security: Protect to Enable. Apress Publishing*. Apress Open.

E-Book in UoP library

Alexander, D., Finch, A., Sutton, D. and Taylor, A., 2020, Information Security Management Principles (3rd Edition), BCS. (ISBN 9781780175188)

Further Indicative Reading

It is expected that students do regular reading and research on the topics each week. There are a range of good security related books available in the library and bookshops, as well as very relevant material online and in research outlets such as journals and conference proceedings.

The Cyber Security Book of Knowledge (v1.1)
https://www.cybok.org/knowledgebase1_1/ (a comprehensive source of up-to-date security knowledge, backed by the NCSC)

Alan Calder (2015), IT governance: an international guide to data security and ISO27001/ISO27002 (Guidance for companies looking to protect and enhance their information security management systems and protect themselves against cyber threats)

Alan Calder (2013), ISO27001 / ISO27002: A Pocket Guide (This pocket guide gives an overview of two of the key information security standards: the ISO 27001 requirements for creating an ISMS, and the ISO 27002 best-practice recommendations for initiating, implementing, and maintaining it.)

Raymond Panko (2009) Corporate Computer and Network Security. Pearson; ISBN-10: 0136121578. (Good all round security book. There is also a 2004 version which covers similar material and the latest version is the 2011 version – it covers mostly the same things as the 2009 version with some updates).

Gurpreet Dhillon (2007) Principles of Information Systems Security. Wiley & Son. ISBN 0-471-45056-1 (Good all-round book on information security – still very relevant, particularly on security processes, for the current security climate).

Ananda Mitra (2010) Digital Security: Cyber Terror and Cyber Security (Digital World). Chelsea House Publishers; ISBN-10: 0816067910. (Focuses more on cyber security – as per the title).

In addition, there is a range of security related books and e-books available from the university library.

Assessment details

This module has two assignments.

Assignment 1 (70%) is completed over the duration of the module and consists of:

- A proposal for a report into Information Security concerns (500 words, 10 %)
- A report into Information Security concerns (2500 words, 50 %)
- An engagement mark based on Moodle participation (10 %)

Assessment 2 (30%) is a 1-hour Moodle quiz and will occur in the TB2 examination period. It will consist of a mix of question styles and cover content from Week 1-11. This exam is an limited open book exam and you can bring in notes that consist of 2 sheets of A4 paper.

Examples and practice questions will be provided during the course of the module and a practice exam provided in Week 11.